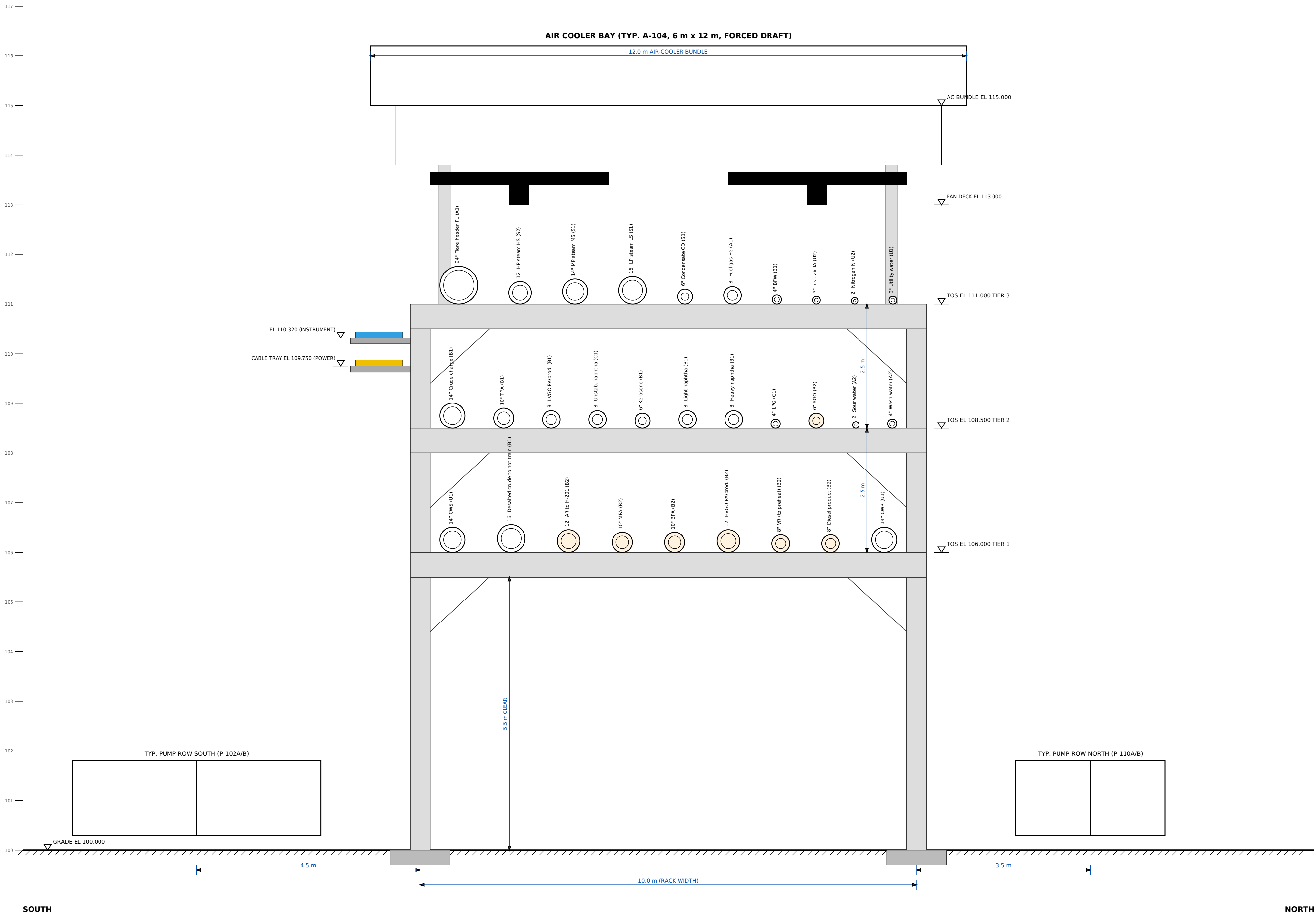


TYPICAL BENT CROSS-SECTION (LOOKING WEST ALONG RACK) - SCALE 1:50

TIER ALLOCATION

Tier	TOS EL	Allocation
1	106.000	Hot process lines (B2/B3), crude, residue, pumparounds
2	108.500	Product, cold HC, sour water, OSBL product lines
3	111.000	Utilities: steam/condensate, CW, FG, N2, IA, flare header
Brkt	109.750	Cable trays EL, power + IC (south brackets)
AC	115.000	Air-cooler bundles A-101..A-202 (10 bays)

Utilities on tier 3 include steam headers with expansion loops;
loops for tier-1 hot lines raised to tier 3 level at loop bays.



- NOTES**
- Typical bent; bents at 6.0 m centres (12 m portal over RD-W). Steel HE400B columns, HE300A beams.
 - Line sizes indicative: from pump rated flow at 2.0 m/s (liquid); final sizes per line list.
 - Hot lines (> 260 C) on tier 1 with expansion loops on tier 3 level; flare header
 - on tier 3 free-draining to D-104 (L-500). CWS/CWR heavy lines tier 1 outer positions.
 - Cable trays (EL / IC) on south cantilever brackets, segregated from hot lines; fireproofed
 - rack steel to EL 111.0 below air coolers per API 2218 (to be confirmed by fire study).
 - Pump CL 3.5 m (north) / 4.5 m (south) off rack edge; drivers accessible from under-rack aisle (ER-1).
 - Headroom under tier 1 = EL 106.0 - beam - road crown >= 5.0 m (no mobile crane under rack).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPE RACK TYPICAL SECTION			
MAIN PIPE RACK PR-100 - TIER ALLOCATION			
DISCIPLINE	STAGE	SCALE	SHEET
PLANT LAYOUT	FEED	1:50	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-PL-ELV-003			A